

Input

Graph Structure
 $\mathcal{G} = (\mathcal{V}, \mathcal{E})$



Node Features
 $\mathbf{X} \in \mathbb{R}^{|\mathcal{V}| \times F}$

v_1						...	
v_2						...	
$\vdots$						...	
$v_{ \mathcal{V} }$						...	

Data Augmentation

View 1



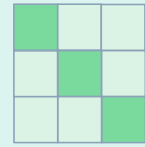
View 2



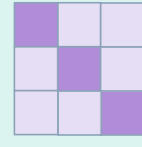
Generate two augmented graph views as the input

CDMD: Cross-Dimensional Mutual Distillation

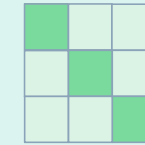
Low- d_i Sim Matrix



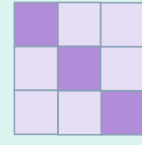
Max-d Sim Matrix



Low- d_j Sim Matrix



Max-d Sim Matrix



: Similarity Matrix $\longleftrightarrow$: Similarity/Alignment

Align similarity distributions with symmetric KL divergence

HPEM: Hard-Pair Evolutionary Mining

Small-d Prefix d_{i-1} $\xrightarrow{\text{Difficulty Evolution}}$ Large-d Prefix d_i

Representation Capacity

Capacity Used Limited

Representation Capacity

Previous Capacity Extra Capacity

Node Difficulty (by confusion)



Hard Samples Discovered



Focus on Hard Samples

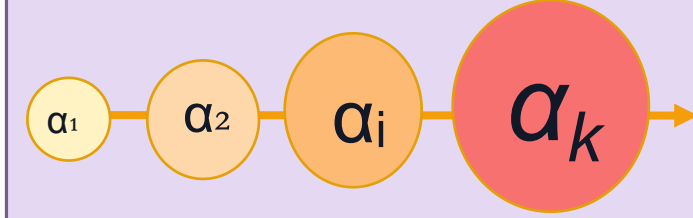


More optimization focus : Higher Weight

Use small-d confusion to weight large-d InfoNCE loss

DALS: Dimension-Adaptive Loss Scheduling

$$\alpha_i = \exp\left(\lambda \frac{d_i}{d_K}\right)$$



Weights increase with dimension:
 $\alpha_1 < \alpha_2 < \dots < \alpha_i < \dots < \alpha_k$

Weighted HPEM Loss Aggregation

$$\alpha_1 \cdot \mathcal{L}_{HPEM}^1 + \alpha_2 \cdot \mathcal{L}_{HPEM}^2 + \dots + \alpha_k \cdot \mathcal{L}_{HPEM}^k$$

Effect: Higher-dimensional prefixes receive larger weights, devoting more optimization resources to learning difficult node pairs.

Joint Optimization:

$$\mathcal{L}_{MCNE} = \lambda_{CDMD} \mathcal{L}_{CDMD} + \frac{1}{K} \sum_{i=1}^K \alpha_i \mathcal{L}_{HPEM}^{(i)}$$